

Question			Marking details	Marks available				Maths	Prac												
				AO1	AO2	AO3	Total														
3	(a)		ΔU increase (or change) in internal energy of a system (1) Q heat flowing into the system (1) W work done by the system (1) Reference to system at least once otherwise award a maximum of 2 marks	3			3														
	(b)	(i)	<table border="1"><thead><tr><th>Process</th><th>Description of process</th><th>Work done on/by gas (if any)</th></tr></thead><tbody><tr><td>A $\rightarrow$ B</td><td>Increase in pressure at constant volume</td><td>No work done (1) – AO2</td></tr><tr><td>B $\rightarrow$ C</td><td>Decrease in volume at constant pressure (1) – AO1</td><td>Work done on gas (1) – AO2</td></tr><tr><td>C $\rightarrow$ A</td><td>[Linear] decrease in pressure with increasing volume (1) – AO1</td><td>Work done by gas (1) – AO2</td></tr></tbody></table>	Process	Description of process	Work done on/by gas (if any)	A $\rightarrow$ B	Increase in pressure at constant volume	No work done (1) – AO2	B $\rightarrow$ C	Decrease in volume at constant pressure (1) – AO1	Work done on gas (1) – AO2	C $\rightarrow$ A	[Linear] decrease in pressure with increasing volume (1) – AO1	Work done by gas (1) – AO2	2	3		5		
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		(ii)	Work done on the gas is given by the "area" enclosed $= \frac{1}{2}(8 - 4)10^{-3}(2 - 1)10^5$ (1) $= \frac{1}{2}(4)(1)10^2 = 200$ [J] (1)		2		2	2													
	(c)	(i)	Use of $PV = nRT$ and note T constant (1) At A $PV = (1 \times 10^5)(8 \times 10^{-3}) = 800$ [J] (or at C $PV = (2 \times 10^5)(4 \times 10^{-3}) = 800$ [J]) (1) Check any point on curve e.g. <table border="1"><thead><tr><th>If V (m³)</th><th>P (Pa)</th></tr></thead><tbody><tr><td>5.0×10^{-3}</td><td>1.60×10^5</td></tr><tr><td>6.0×10^{-3}</td><td>1.33×10^5</td></tr><tr><td>7.0×10^{-3}</td><td>1.14×10^5</td></tr></tbody></table> (1)	If V (m ³)	P (Pa)	5.0×10^{-3}	1.60×10^5	6.0×10^{-3}	1.33×10^5	7.0×10^{-3}	1.14×10^5	1	1		3	3					
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		(ii)	Attempt at calculation of area – [accept small square count of between 170 and 280 i.e. $225 \pm \sim 25\%$] (1) If 225 i.e. 2.25 large squares, work difference = $2 \Delta V \Delta p$ (ΔV and Δp for large square) $= 2.25 (1 \times 10^{-3})(0.2 \times 10^5) = 45 \text{ J}$ with uncertainty of 25%, accept a value between 34 and 56 J provided method correct (1)			2	2	2	
			Question 3 total	6	7	2	15	7	0